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Machine Learning Based Automated Pressure Testing in Carbonate Formations of UAE

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Abstract

Wireline formation pressure tests (pretests), are critical in reservoir characterization during exploration stage, providing reservoir pressure and mobility. We have demonstrated the application of a novel and fully automated digital pretest workflow in onshore, UAE. The use of intelligent real-time pretest classification combined with automated setting and retract of pretest probes led to significant reduction in overall test durations with increased consistency in results. These advancements allowed the reservoir engineers and asset managers to focus on interpretation of results, enhancing reservoir characterization and reducing operational risks

This novel workflow is enabled by four automation stages combined and executed over a single command from a new customized acquisition system at wellsite: Automated setting of pretest probes, automated real time pretest classification (RTPC), automated pressure and mobility estimation and automated retract of probes. The new acquisition system is mounted with both logging tool electronic control algorithm and pretest interpretation algorithms, hence bringing the pretest interpretation to the wellsite minimizing the time loss due to delay in signal transmission from wellsite to the office. The first stage includes automated setting of probes in less than 30 seconds compared to conventional duration of 90 seconds. Next stage is automated pretest classification, where K-nearest neighbor (KNN) & Light grading boosting machine (LGBM) based AI/ML algorithms are used to classify the pretest type as valid test or invalid test. For valid tests, pressure stabilization is monitored over 10 second windows, and mobility is auto computed. For an invalid test, the fourth automation stage is triggered, leading to retracting probes in 10 seconds with 95% accuracy. This process reduced manual interventions from 25 clicks to 5 per station, ensuring consistency in operations across multiple depths.

A total of 93 pretests were taken using a pre-planned automation algorithm executed over single command at each depth. The improvement in pretest efficiency and consistency was measured in time saving and data quality. Around 80% reduction in time spent on probe setting and retract was observed. The average pretest station time reduced from 13 minutes to less than 8 minutes, achieving a 38% overall time reduction. Tight tests and valid tests saw a 50%- and 40%-time reduction, leading to an overall time saving of over

7 hours during the pretest logging operation. Faster testing in openhole conditions also reduced wireline sticking risks.

We showcased the application of AI & ML automation algorithms, tested and trained over 890 and 153 datasets respectively, and deployed on more than 20 real time pretest operations across the globe, before being used in this pretest operation. The real time pretest classification (RTPC) brought speed and consistency by reducing the coefficient of human error in such long pretest operations, leading to an efficient and standardized way of pressure testing.

Introduction

Formation testing logging services play a vital role in the measurement of reservoir pressure and mobility. They are often executed via a wireline conveyed openhole logging tool, consisting of a probe for establishing communication with the formation, a downhole pressure gauge and hydraulic assembly to enforce pressure drawdown. A key advantage of formation testing is its speed of pressure measurement and can be used to take consecutive or multiple pressure tests at different depths in an open hole. These measured formation pressures can be used to construct a formation pressure gradient, which can provide an initial idea about the possible reservoir fluid across the formation. Multiple gradients can later lead to fluid contacts derivation. Pressure gradients also provide insights into reservoir layer connectivity based on differences in pressure regimes.

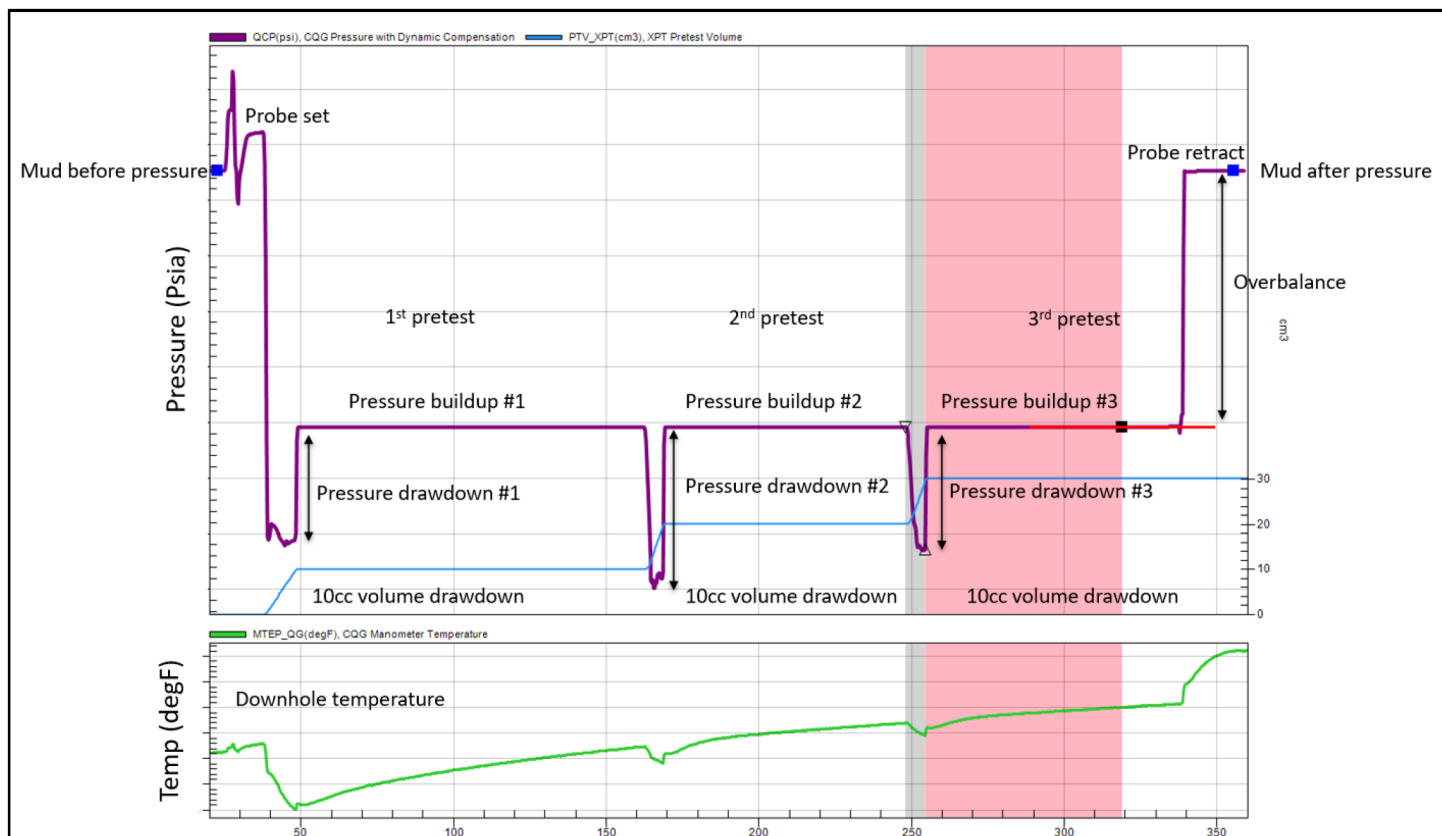


Figure 1—Typical pressure test (pretest) plot- pressure channel (in magenta) and temperature (in green)

Figure 1 above showcases a typical pressure test (pretest) sequence showcasing the key steps involved, as described below:

1. The pretest probe is lowered to the desired depth of pressure measurement, and at this depth the downhole pressure gauge in probe assembly reads the hydrostatic mud column pressure (also referred to as mud before pressure in [figure 1](#)).
2. The probe set sequence is started, and the probe pierces across the mud cake and makes a seal with the formation wall. A surrounding concentric packer further maintains this seal.
3. The initial drawdown sequence is initiated at a defined volume and rate, and a drop in downhole gauge pressure is observed. At this stage, also flowline decompression takes place.
4. Once the drawdown is completed, a pressure buildup is observed. If the seal between formation and probe is intact, the pressure builds up and stabilizes at the formation pressure levels. The pressure buildup stabilization is monitored by a parameter known as 60 sec slope (psia/min), which eventually reaches towards zero.
5. The drawdown-buildup sequence is repeated multiple times, to get the repeatability in pressure buildups.
6. Once stable and repeated formation pressure measurements are achieved, the probe is retracted.
7. The probe is then moved to the next desired position for pressure measurements.

In the above process, a lot of manual interventions are required. A group of engineers including reservoir engineers and tool operators monitor the pressure test in real time and do required interventions by giving commands for taking additional drawdowns or checking the overall quality of test- whether it is giving a valid formation pressure or not. With the advancements in digital technology and use of machine learning, we have devised a way to automate this entire process to bring efficiency, speed and accuracy to the whole process. The first level of automation was to define the overall validity of a pretest and do the macro classification as a valid test or an invalid test. An invalid test can be a test with lost seal / no seal, tight test and dry test, as shown below in [figure 2](#), with the help of such examples of invalid tests.

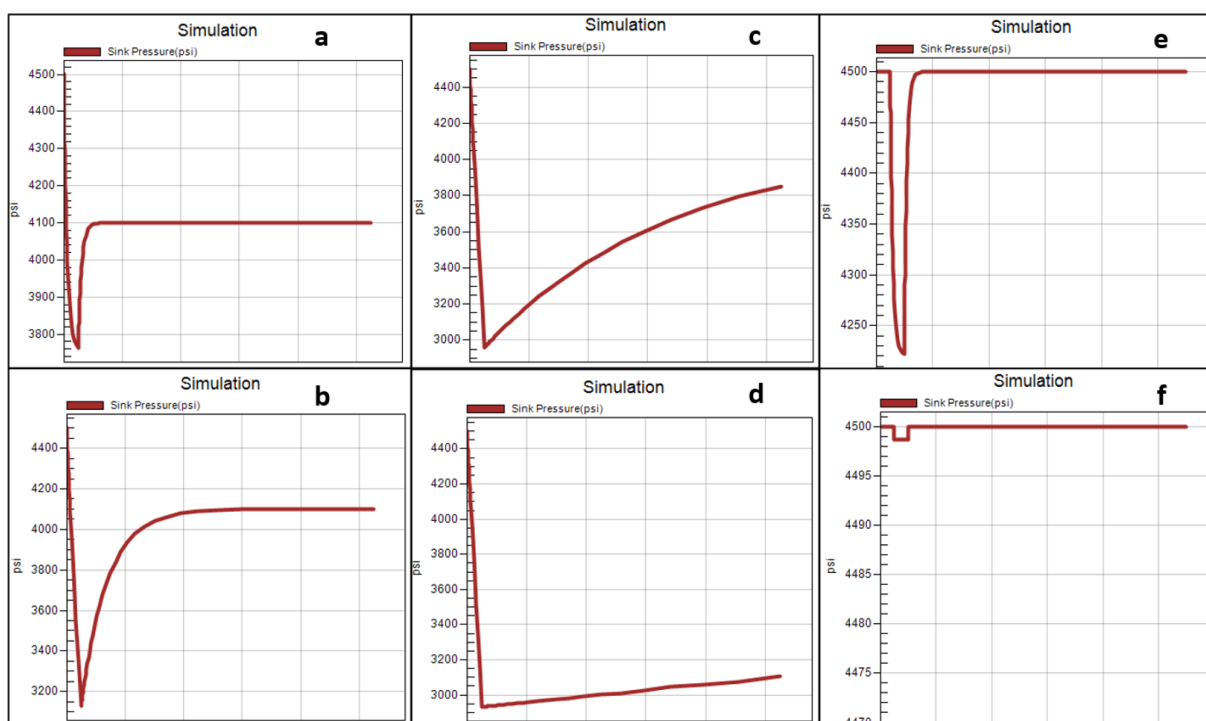


Figure 2—pretest classification illustration through simulated pretests: a) valid test for moderate to good mobility, b) valid test with low mobility, c) Invalid test- tight test, d) Invalid test- dry test, e) Invalid test- Lost seal during drawdown f) Invalid test- No seal

As shown above in [figure 2](#), the slope of pressure buildup curve is used for classification of a valid test or invalid test (tight test or dry test). A tight test stabilizes over long periods of time compared to a valid test with ranges of mobility. However, a dry test doesn't attain the trajectory of buildup, hence doesn't get to stabilize. A lost seal is a test where a valid drawdown is observed as seen in [figure 2\(e\)](#), however the seal was lost, and pressure went back to hydrostatic. In a no seal, from start of drawdown, no seal between formation and probe was observed. The classification of pretests is manually done in real time by examining the pressure buildup and 60 sec slope (psia/min), followed by command to take additional drawdowns. To automate this process, we have developed a machine learning algorithm for auto-identification of pretests as valid or invalid tests. The ML algorithm analyzes the shape / curvature ([Sarili et al., 2024](#)) of pressure buildup curve to compute the 60 sec stabilization slope after every 10 seconds and auto-identify the nature of pretests. If a test is valid, it calculates the mobility of pretest and accordingly provides the designed volume and rate for next pretest, and additionally it sends the command to acquisition system for initiating the pressure drawdown sequence. In the past, the pretest parameters have been optimized ([Loviken et al., 2024](#)), based on the response of initial drawdown and buildup. In our automated algorithm, we have given flexibility for users to pre-decide the parameters for valid tests with mobility ranges (very low mobility, low mobility, medium mobility, high mobility) as well as for invalid tests like tight test, dry test, lost seal and no seal.

Description of workflow

A machine learning algorithm was developed based on analyzing the theoretical base of pretests, combining the reservoir parameters as well as tool parameters for enhancing and automating the pretest classification and monitoring decisions in real-time. The workflow uses this algorithm for auto-identification of pretest type, as well as take necessary commands to proceed i.e., take additional pretests or retract probe. [Figure 3](#) below shows the workflow steps from openhole log acquisition till completion of pretest operation. Based on open hole log analysis, pretest depths are selected across the open hole. Also, pretest plan is prepared, which includes pretest volume and rate parameters, stabilization parameters, total test duration for different pretest types- valid test with high mobility, medium mobility, low mobility, extra-low mobility, tight test, dry test, lost seal and no seal. Early detection and classification of pretests help to reduce the station time and take decisions in real time. The automation workflow allows users to modify the parameters in the middle of the test, in case of anomalies encountered.

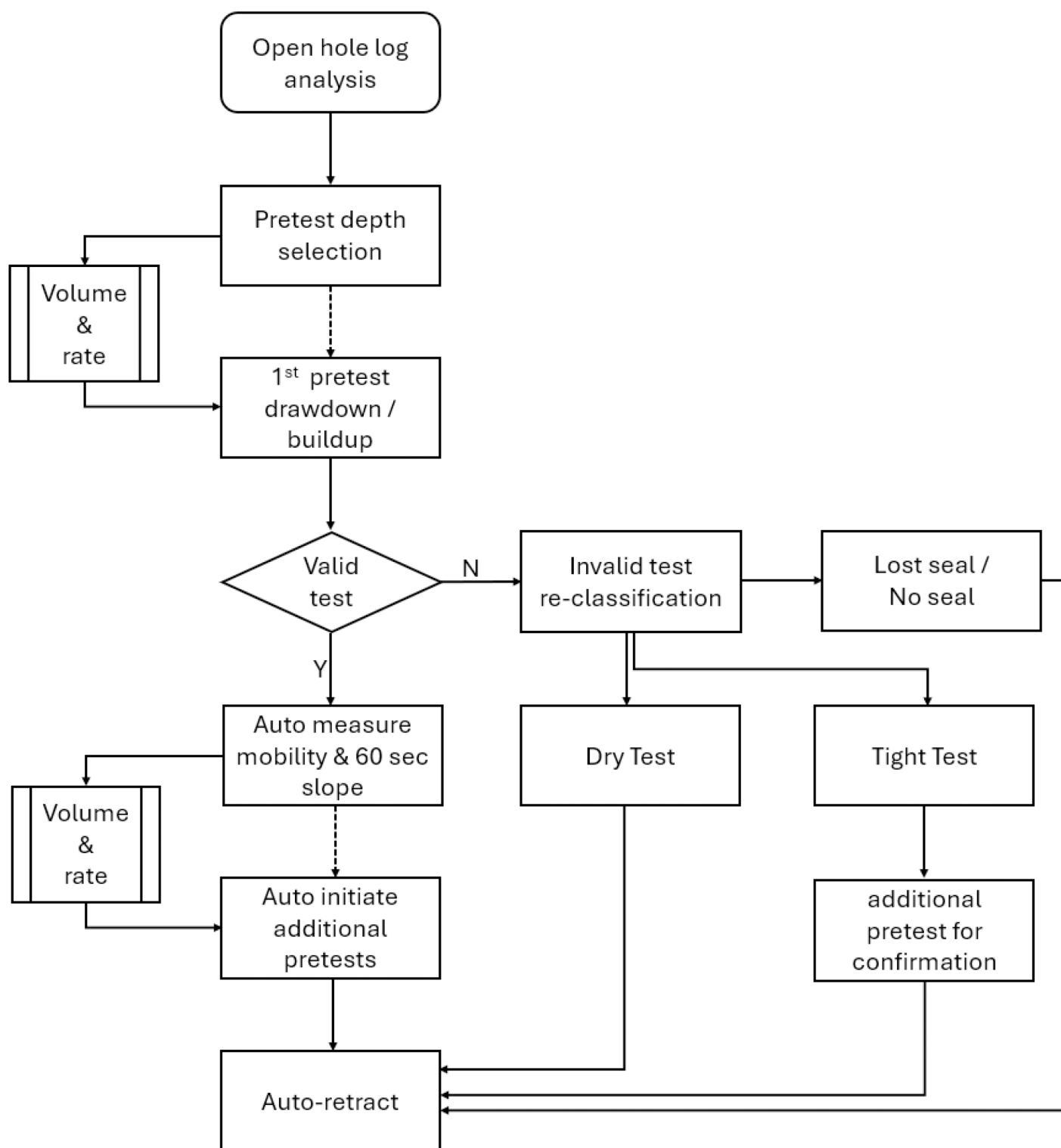


Figure 3—Auto pretest workflow, customizable based on factors like test duration and lithology

Machine learning algorithm

The machine learning system was engineered using K-nearest neighbor (KNN) and Light Gradient Boosting Machine (LGBM) for automated categorization. It begins by detecting pressure test markers at the start and end of pressure buildup, followed by generating a two-dimensional pretest graph, culminating in automated classification. Validation steps are implemented to address erroneous test classifications by analyzing pretest

buildup patterns. Pattern similarity is quantified using the KNN approach, a common technique in data categorization. Labelled datasets predict the class of new entries. A new entry's class is determined by a majority vote among its nearest neighbors, assigned to the dominant class among its K nearest neighbors in the dataset (K being a positive integer). For instance, if $K = 1$, the entry is classified identically to its closest match. Proximity is measured by computing the distance between the new entry and dataset entries, typically using Euclidean distance, though other metrics may apply. Neighbor votes can be equal or weighted by proximity, giving closer neighbors more influence. A key consideration is choosing K , which depends on the dataset. Larger K values reduce noise effects but may obscure class boundaries. Optimal K is determined by testing various values and evaluating model performance. For a given pressure buildup curve, we compute the ratio of standard tests among neighboring curves, transforming pattern similarity into a numerical value.

Beyond curve analysis, pressure buildup stability is influenced by physical parameters: fluid volume, drawdown mobility, flow rate, first and second derivatives at 30 and 60 seconds, 60-second slope, statistical variance, and signal noise. These parameters, combined with a numerical similarity metric, serve as input features for the machine learning model to replicate expert judgment. The LGBM algorithm performs the final classification using these inputs. LGBM employs a decision tree structure, where internal nodes assess features, branches represent evaluation outcomes, and leaf nodes denote final class labels. In tree-based machine learning, labelled datasets automatically construct the tree. Tree-building algorithms work top-down, selecting the feature that best segments the data at each step, often using criteria like the Gini index to measure subset uniformity. Decision trees are popular in machine learning due to their insensitivity to feature scaling, robustness to irrelevant features, and interpretable models, with visualizable decision processes. However, deep trees may overfit, capturing noise in the training data. Gradient boosting mitigates this by combining multiple deep trees trained on data subsets, reducing overfitting. This sacrifices some precision and interpretability but improves overall performance. Unlike single trees, which are noise-sensitive, averaging multiple uncorrelated trees enhances robustness. Gradient boosting uses a unique tree-building method, randomly selecting feature subsets at each split, known as "feature boosting," to avoid over-reliance on dominant features, reducing tree correlation and boosting model resilience. For practical applications, the algorithm provides a binary classification and a numerical confidence score. For a new pressure test, higher confidence is assigned if similar tests exist in the training data. Here, LGBM's output, ranging from 0 (tight test) to 1 (normal test), serves as the confidence measure.

The system was developed and validated using two different wireline tools: one electromechanical, one hydraulic, performing reliably for both. However, field heterogeneity limits the generalizability of a single trained model. Identical pressure tests may be interpreted variably by engineers, especially due to reservoir properties. In low-permeability reservoirs, a "tight" test might be deemed normal. Thus, tight/normal classifications must be tailored to the field's expected mobility. The approach involves training distinct models for fields with varying mobility, using their specific pressure data. The resulting ML system improves data annotation and preparation, enhancing training efficiency.

Case study

Well A was drilled with WBM as an onshore vertical well, upto total depth X800 ft TVD in onshore oilfield, United Arab Emirates. Upon reaching TD, wireline open hole logs were acquired including gamma ray, resistivity, density, porosity, acoustic. Based on open hole log data, multiple low resistivity pay zones were identified. Pressure test plan was prepared to measure formation pressure, mobility and gradients for fluid gradient / contact estimation. A total of 83 depths were selected for pretests lying across various sand zones. Based on the open hole log responses, pretest parameters were defined for valid tests (extra low mobility, low mobility, medium mobility, high mobility), tight tests, dry tests, lost seals and no seals. [Table 1](#) below showcases the pre-planning of pretest parameters, which could be optimised during the pretest operation by reservoir engineers based on specific pretest's responses.

Table 1—Pre-planned pretest parametric table (input to the automatic pretest algorithm)

Mobility	Decision term	Pretest volume (cc)	Pretest rate (cc/sec)	60 sec slope (psia/min)	Repeatability (psia)	Maximum test duration (min)	Drawdown pressure limit (psia)
x-low	Initial pretest	5	0.3	1	10	15	Hydrostatic - 1000
x-low	Repeat pretest	5	0.3	1	10	15	Previous DDPL
low	Initial pretest	5	0.5	0.1	0.1	12	Hydrostatic - 1000
low	Repeat pretest	5	0.5	0.1	0.1	12	Previous DDPL - 500
medium	Initial pretest	10	1	0.05	0.02	10	Hydrostatic - 1000
medium	Repeat pretest	10	1	0.05	0.02	10	Previous DDPL - 500
high	Initial pretest	10	2	0.01	0.01	8	Hydrostatic - 1000
high	Repeat pretest	10	2	0.01	0.01	8	Previous DDPL - 500

In above [table 1](#), the ranges of mobility as extra-low, low, medium or high are pre-defined as given in [table 2](#).

Table 2—mobility range definition table

Mobility type	Range (mD/cP)
X-Low Mobility Range	$0 < \text{mobility} < 0.3$
Low Mobility Range	$0.3 < \text{mobility} < 3$
Medium Mobility Range	$3 < \text{mobility} < 10$
High Mobility Range	$\text{mobility} > 10$

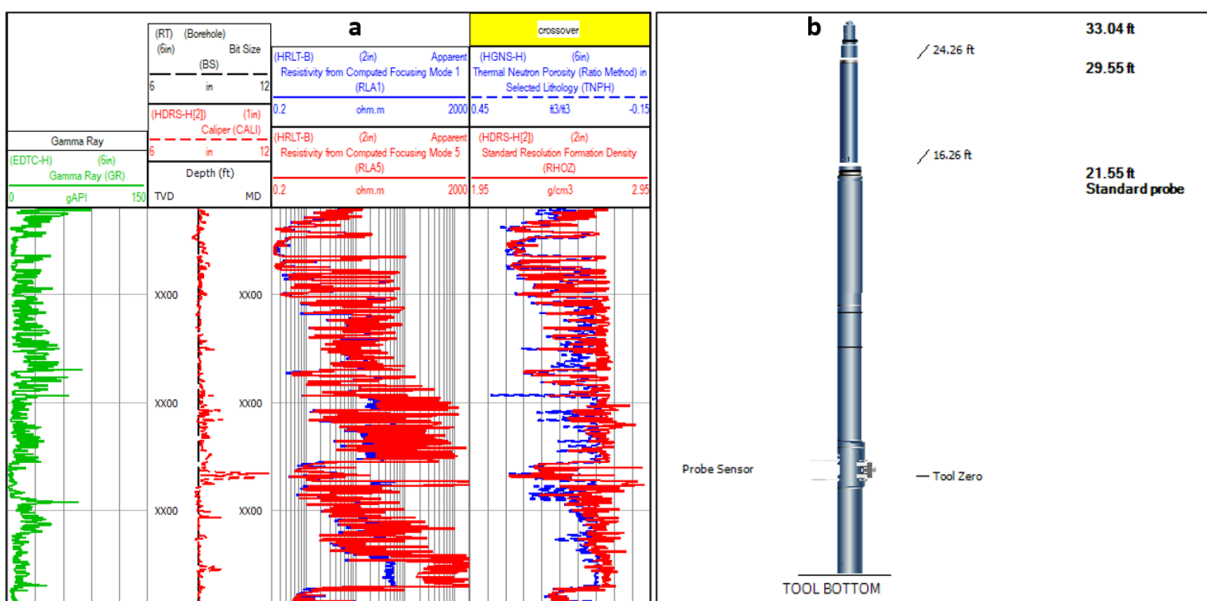


Figure 4—a) open hole logs: track 1- gamma ray, track 2- bit size, caliper, depth, track 3- shallow and deep resistivity, track 4- density, porosity logs b) pressure test tool string

Pretest cases

Pretest example #1: Based on open hole log response, a medium mobility range pretest was initiated at xx48 ft MD. During the first pretest, a mobility of 2.23 mD/cP was observed, which lies in low mobility regime. The automated pretest algorithm computed the mobility in real time and switched the pretest parameters for second pretest to a low mobility parameter case. This decision to change the pretest volume and rate parameters in real-time was taken by the ML algorithm, reducing the decision-making time and augmenting the reservoir engineer as well as wireline field engineer's decisions. Figure 5(a) shows the three pretests taken automatically using the ML algorithm based auto pretest workflow, which executed entire spectrum of pretesting like setting probe, taking first pretest, measuring its stability, calculate the mobility and decide its range from table 2, decide on second pretest volume and rate parameters as per table 1, take the second pretest, repeat the sequence till the total pretest time reaches the limit provided as per table 1.

Pretest example #2: Based on open hole log response, a medium mobility range pretest was initiated at xx98 ft MD. During the first pretest, a pressure drop of #1000 psia was observed. The pressure buildup shape was evaluated by ML algorithm as tight test, and based on stabilization rate auto-pretest algorithm executed the auto retract mechanism. The entire test was finished in less than 2 minutes, compared to a normal average of 6-7 minutes (around 70% of the time saved in invalid test), as shown in figure 5(b).

Pretest example #3: Based on open hole log response, a low mobility range pretest was initiated at xx79 ft MD. During the first pretest, a pressure drop of #2000 psia was observed. The pressure buildup shape was evaluated by ML algorithm as dry test, and based on stabilization rate auto-pretest algorithm executed the auto retract mechanism. The entire test was finished in less than 1.5 minutes, compared to a normal average of 6-7 minutes (around 80% of the time saved in invalid test), as shown in figure 5(c).

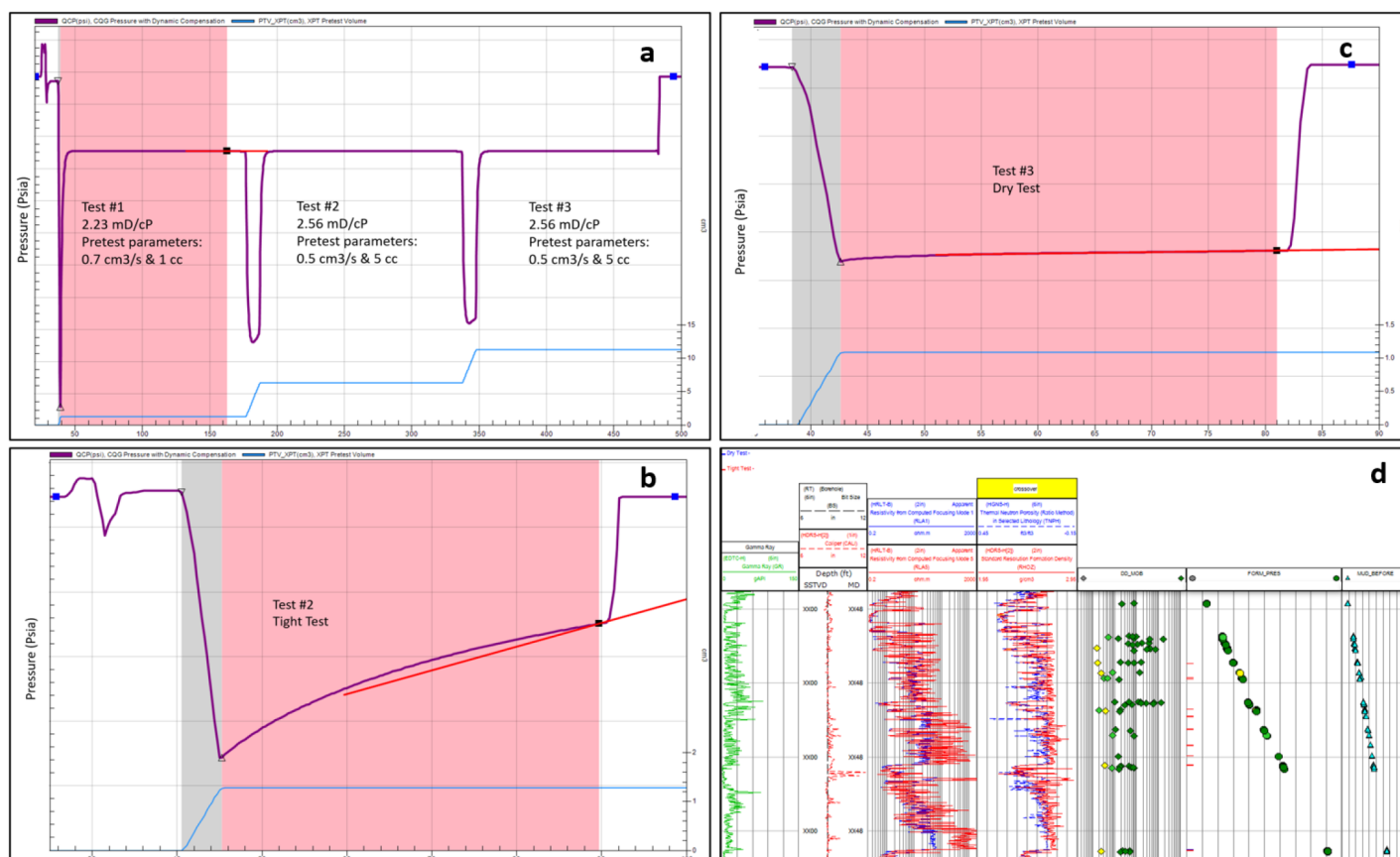


Figure 5—a) pretest #1, b) pretest #2, c) pretest #3, d) 2D depth view: track 1- gamma ray, track 2- bit size, caliper, depth, track 3- shallow and deep resistivity, track 4- density, porosity logs, track 5- mobility, track 6- formation pressure, track 7- mud before (hydrostatic pressure).

Conclusion

In this paper, we demonstrated the machine learning based auto-pretest algorithm for automated real time execution of pretests. The algorithm helps in faster decision making by auto-classification of pretest type and execution of secondary pretests, measure its stability and repeatability. All these steps were manually executed by reservoir engineers and wireline field engineers, however with the help of automated algorithms, it provides more time for users to focus on pressure quality, real time gradient construction and taking faster decisions. A special feature and advantage of this algorithm / workflow is minimal time spent at an invalid test station, saving critical rig time. Figure 6 below showcases a comparison on test durations from automated pretest algorithm versus manual pretest approaches. A total of 93 pretests were acquired using auto-pretest algorithm, with an overall average station time of #7.7 min/station, compared to #12.4 min/station using conventional manual approach, led to a saving of over 9 hours of rig time and resources including CO₂e emissions from running rig setup. For invalid tests (tight tests in this case), more than 50% reduction in station time was observed.

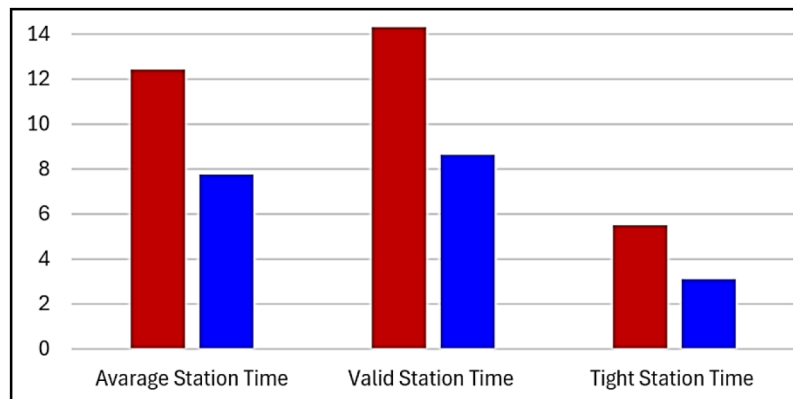


Figure 6—station time comparison plot: red bar showcases the station time using conventional manual approach, blue bar represents station time using automated pretest algorithm.

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